

1. Administrative & Study Identification

Study Title: A Clinical Study of MK-2870 Alone or With Other Treatments to Treat Gastrointestinal Cancers (MK-9999-02A)
Official Title: A Phase 1/2 Study to Evaluate the Safety and Efficacy of MK-2870 Monotherapy or in Combination With Other Anticancer Agents in Gastrointestinal Cancers **Short Title/ Acronym:** LIGHTBEAM-02A **Protocol Number:** MK-9999-02A **NCT Number:** NCT06428409 **Posting ID:** 695125335035779127 **Trial Status:** RECRUITING **Sponsor / Company:** Merck (Merck Sharp & Dohme LLC)
Investigational Product: Sacituzumab tirumotecan (MK-2870 / sac-TMT) **Phase of Development:** PHASE1, PHASE2 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and adverse event profile of sacituzumab tirumotecan (sac-TMT) alone or in combination with other treatments in gastrointestinal cancers. **Secondary Objectives:** To evaluate Duration of Response (DoR), Progression-Free Survival (PFS) per RECIST v1.1 by Blinded Independent Central Review (BICR), and Overall Survival (OS).

2.2 Background & Rationale There is a significant unmet medical need for new, tolerable, and effective therapeutic options for colorectal cancer, pancreatic ductal adenocarcinoma, and biliary tract cancer that can provide a survival benefit. TROP2 is a cell surface antigen broadly expressed in several gastrointestinal cancers. Sac-TMT (MK-2870) is an investigational antibody-drug conjugate (ADC) designed to deliver targeted tumor cell destruction.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm/Parallel Assignment (Multi-cohort) **Blinding:** Open-label **Randomization:** Non-randomized **Trial Status:** RECRUITING

3.2 Planned Enrollment & Duration Targeted Enrollment: 180 evaluable participants **Number of Sites:** Global multi-center (including US, Australia, and UK sites) **Estimated Study Start:** 20-Jun-2024 **Estimated Primary Completion:** 16-Oct-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Unresectable or metastatic colorectal cancer and has received prior therapy for the cancer. 2. Advanced or metastatic pancreatic ductal adenocarcinoma (PDAC) and has received prior therapy for the cancer. 3. Advanced and/or unresectable biliary tract cancer (BTC) and has, or has not, received prior therapy for the cancer. 4. Has recovered from any side effects due to previous cancer treatment (for participants who have received prior therapy).

4.2 Exclusion Criteria 1. History of severe eye disease. 2. Received prior systemic anticancer therapy including investigational agents within 4 weeks before starting study intervention (for participants who have received prior therapy). 3. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Dose-Escalation/Combo (Cohort 1 - CRC):* sac-TMT 3 mg/kg or 4 mg/kg IV Q2W (Days 1 and 15 of 4-week cycle) + 5-FU 2400 mg/m² IV over 46-48 hours Q2W + Leucovorin 400 mg/m² IV Q2W. * *Monotherapy (Cohorts 2/3 - PDAC/BTC):* sac-TMT 4 mg/kg IV Q2W (Days 1 and 15 of 4-week cycle). * *Other Combination Agents:* Other treatments may include cisplatin (Trade Name: Platinol, ATC Code: L01XA01) depending on specific cohort evaluations. **Route of Administration:** Intravenous (IV) Infusion **Duration of Treatment:** Up to approximately 2 years (approx. 27 rounds) of treatment, followed by up to 3 years of follow-up (5 years total).

5.2 Concomitant & Prohibited Therapy Permitted: Rescue medications (antihistamines, H₂ receptor antagonists, acetaminophen, dexamethasone) for infusion/hypersensitivity reactions. Supportive care measures including antidiarrheals, antiemetics, and artificial tears/gels for ocular surface toxicity. **Prohibited:** Concurrent investigational agents and strong immunosuppressive treatments outside of protocol guidelines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Prior Therapy Review, Baseline Imaging
Baseline	Day 1 (Cycle 1)	First Dose IV Infusion, PK/Safety Blood Draws, Vitals
Interim	Q2W (Days 1 & 15)	IV Infusion, Safety Labs, Efficacy Assessment (RECIST v1.1),

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Number of Participants Who Experience a Dose-limiting Toxicity (DLT):** A DLT is a medical problem related to the study medicine that prevents giving participants a higher dose or may prevent giving the participant the same dose.
- 2. Number of Participants Who Experience One or More Adverse Events (AEs):** An AE is a health problem that happens or worsens during the study.
- 3. Number of Participants who Discontinue Study Treatment due to an AE:** The number of participants who stop study treatment due to a health problem that happens or worsens during a study.

7.2 Secondary Outcomes

- 1. Duration of Response (DOR) per RECIST 1.1 as Assessed by BICR.**
- 2. Progression-free Survival (PFS) per RECIST 1.1 as Assessed by BICR.**
- 3. Overall Survival (OS).**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for dose-limiting toxicities (DLTs), specifically focusing on infusion reactions, ocular surface toxicity, and pneumonitis/ILD. **Serious Adverse Events (SAEs):** Routine 24-hour reporting protocols for severe AEs requiring hospitalization or posing life-threatening risks. **Data Monitoring Committee (DMC):** Internal and external safety review committees to guide dose-escalation phases and monitor overall trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose) and Intent-to-Treat (ITT) population for efficacy. **Sample Size Justification:** Target enrollment of 180 participants is adequately powered to assess safety/DLTs in the escalation phase and estimate ORR bounds in the expansion cohorts across 3 distinct GI cancer types. **Handling of Missing Data:** Standard oncology methods; PFS and OS utilizing Kaplan-Meier estimates and right-censoring for patients lost to follow-up or without events.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite and remote monitoring

corresponding to Q2W infusion cycles. **Audit Trail:** Validated eCRF platforms with electronic data capture, strict RECIST v1.1 image upload compliance, and query resolution tracking.

11. Study Identifiers & Governance

Posting ID: 695125335035779127 **Primary ID:** 9999-02A **Registry Number:** NCT06428409 **Company / Sponsor:** Merck **Study Title:** A Clinical Study of MK-2870 Alone or With Other Treatments to Treat Gastrointestinal Cancers (MK-9999-02A)

12. Clinical Framework (Gastrointestinal Cancer Specific)

Disease Areas: Colorectal Cancer, Pancreatic Ductal Adenocarcinoma, Biliary Tract Cancer **Preferred UMLS Name:** Gastrointestinal Neoplasms **Diagnosis Threshold:** Unresectable, advanced, or metastatic disease **Medical Methodology:** TROP2-directed Antibody-Drug Conjugate (ADC) combined with systemic treatments or as monotherapy **Optimization Target:** Tolerability (AEs/DLTs), Objective Response, and Progression-Free Survival

13. Study Procedures & Methodology

Management Pathway: ADC and Systemic IV Infusion Clinical Pathway **Education Protocol (HCP):** Dose modification guidelines, severe ocular toxicity management, and infusion-reaction protocols **Education Protocol (Patient):** Strict reporting guidelines for respiratory symptoms (pneumonitis risk) and ocular side effects **Quality Audit Mechanism:** Independent central imaging review (BICR) to confirm RECIST v1.1 compliance

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years (approx. 2 years active treatment + 3 years follow-up) **Onsite Visit Cadence:** Every 2 weeks (Q2W) on Days 1 and 15 of each cycle **Remote Monitoring Frequency:** Continuous survival follow-up post-treatment **Checklist Review Interval:** Every 4 weeks (End of each treatment cycle)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				